Supporting Information: Supporting Information: Target breadth and mutation resilience in African-natural-product-inspired antimalarial chemotypes: a computational analysis

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This supplement contains the numerical material needed to reproduce the integrated computational analysis. It is organized by evidence layer: chemical space, target-anchored docking, RRS, and exploratory cross-metric analysis. The four-target matrix is reported target by target; no cross-target mean is used because the Vina score is not calibrated across unlike receptor pockets.

S1. Cohort and evidence boundaries

The integrated analysis uses the 17-member P2 Set-C polypharmacology-oriented cohort. Set C was selected using weighted MPO, synthetic accessibility, QED, and target-profile criteria; all 17 entries had $n_{\text{targets bound}} = 2$ in the source selection table. This enrichment is intentional and limits prevalence-based interpretation. The upstream $n_{\text{targets bound}} = 2$ label describes the P2 selection criterion; it does not constrain the separate 17-by-4 wild-type docking profile reported here. The docking calculations follow AutoDock Vina conventions.[S1,S2] The MPO top-20 and the P2 MD top-20 are distinct sets. The P2 MD systems 201, 438, the 164-labelled PfClpR cohort, and 214 are not MD validation of Set C. The target panel contains four wild-type docking profiles, whereas RRS is calculated from the separate WT/mutant docking panel and is available only for PfDHFR and PfCRT mutant panels.

S2. Chemical-space coverage and target-anchored docking

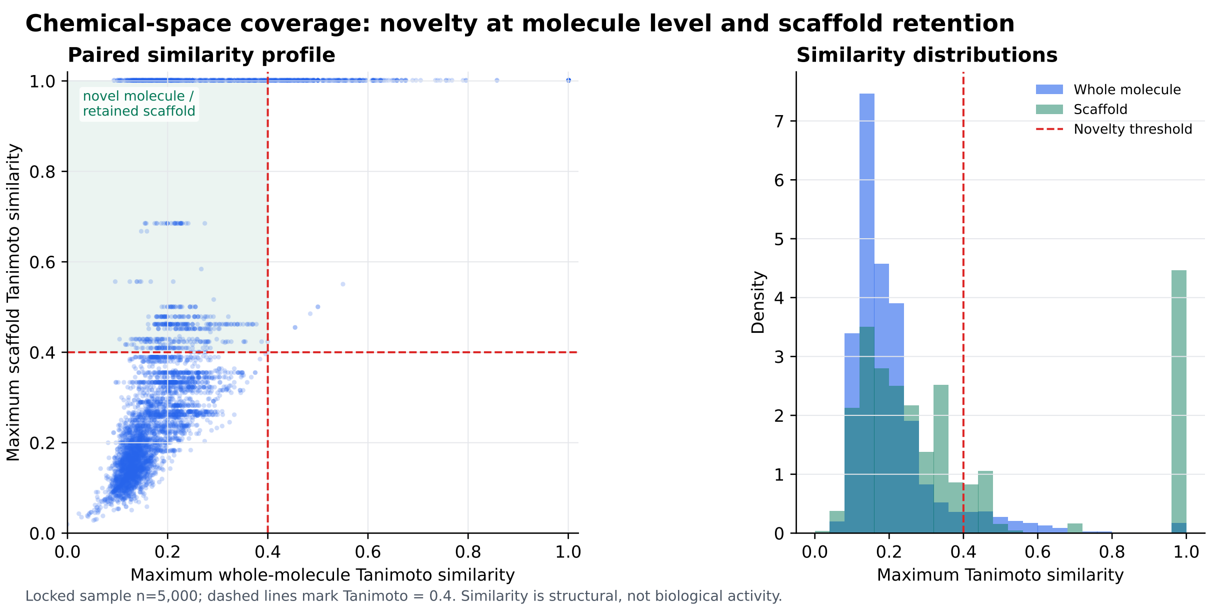

Figure S1: Chemical-space coverage in the locked 5 000-molecule novelty-analysis sample. The paired plot compares maximum whole-molecule and scaffold Morgan-fingerprint Tanimoto similarities to the African-natural-product reference set; the dashed lines mark the operational threshold of 0.4. The upper-left region represents molecules that are dissimilar at the whole-molecule level while retaining scaffold-level similarity. Similarity is a structural descriptor and does not indicate biological activity, target preference, or synthetic success.

S3. Target-anchored docking matrix

Table S1: Target-anchored AutoDock Vina score estimates for the locked, intentionally enriched 17-member Set-C cohort. Values are in kcal mol^{-1}; they are scoring-function outputs, not experimental free energies. Columns are reported target by target because scores from unlike receptor pockets are not calibrated for arithmetic comparison.

Candidate	PfDHFR	PfCRT	PfClpP	PfATP4
PP-01	−5.947	−6.483	−5.467	−6.361
PP-02	−5.562	−6.124	−5.907	−6.591
PP-03	−6.089	−6.633	−6.358	−7.311
PP-04	−5.493	−5.329	−5.386	−6.086
PP-05	−6.285	−6.412	−6.466	−6.378
PP-06	−6.267	−7.907	−7.031	−7.285
PP-07	−6.226	−6.633	−6.226	−5.775
PP-08	−4.935	−5.213	−5.052	−5.494
PP-09	−5.693	−6.871	−5.613	−6.078
PP-10	−6.346	−6.395	−6.016	−6.670
PP-11	−6.179	−6.376	−6.583	−6.724
PP-12	−5.248	−6.357	−5.386	−4.635
PP-13	−5.921	−5.685	−6.454	−7.565
PP-14	−4.863	−5.328	−5.047	−5.460
PP-15	−6.045	−6.084	−6.366	−7.016
PP-16	−5.469	−6.202	−6.203	−6.451
PP-17	−5.266	−5.116	−5.342	−5.360

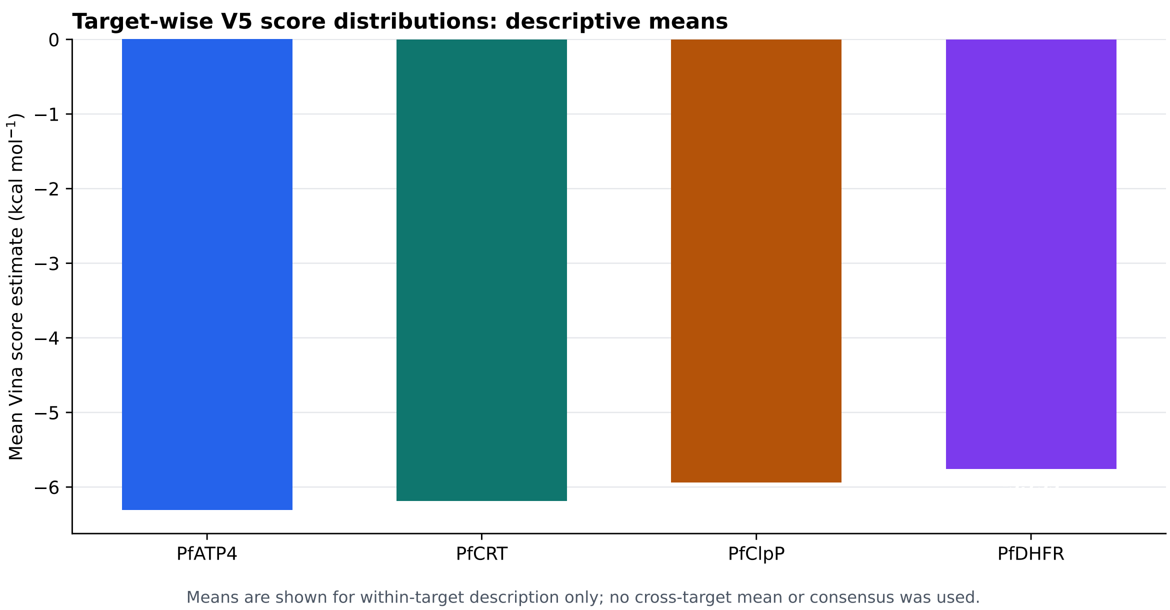

Figure S2: Descriptive target-wise means from the raw docking panel. Bars show the within-target mean across the 17 candidates. Means are shown only within target; no cross-target arithmetic mean or consensus score was used. These summaries are computational scoring-function outputs, not experimental affinity measurements.

S4. Physicochemical properties of Set-C candidates

Table S2: Physicochemical properties of Set-C polypharmacology candidates (PP-01 to PP-17). MW: molecular weight (Da); logP: octanol-water partition coefficient; HBD: hydrogen bond donors; HBA: hydrogen bond acceptors; TPSA: topological polar surface area (Å^2); RB: rotatable bonds; QED: quantitative estimate of drug-likeness (0-1 scale); SA: synthetic accessibility (1 = easy to synthesize, 10 = difficult); Lipinski: number of Lipinski Rule of Five violations.

Candidate	MW	logP	HBD	HBA	TPSA	RB	fsp^3	QED	SA	Lipinski
PP-01	328.3	3.19	1	6	78.1	4	0.17	0.79	2.2	0
PP-02	388.5	3.09	1	6	66.4	8	0.45	0.75	3.3	0
PP-03	356.4	3.38	4	6	107.2	3	0.25	0.63	3.4	0
PP-04	259.3	3.01	0	5	53.7	3	0.21	0.72	2.5	0
PP-05	284.3	2.88	2	5	79.9	2	0.06	0.76	2.1	0
PP-06	354.4	4.32	2	5	76.0	4	0.29	0.80	3.1	0
PP-07	230.3	3.14	0	3	39.4	3	0.21	0.60	2.2	0
PP-08	222.2	2.16	0	4	36.9	4	0.33	0.73	2.6	0
PP-09	185.3	3.68	1	0	15.8	2	0.23	0.69	2.4	0
PP-10	242.2	2.66	1	4	59.7	1	0.07	0.67	2.0	0
PP-11	330.3	2.59	3	7	109.4	3	0.12	0.68	2.4	0
PP-12	222.4	4.40	1	1	20.2	7	0.60	0.63	3.5	0
PP-13	330.3	2.05	3	7	113.3	2	0.18	0.72	3.3	0
PP-14	180.2	2.14	2	3	49.7	2	0.20	0.69	2.3	0
PP-15	316.3	2.03	3	6	96.2	3	0.24	0.80	3.4	0
PP-16	228.2	2.07	1	4	67.5	0	0.08	0.64	2.6	0
PP-17	164.2	1.49	2	2	57.5	2	0.00	0.65	1.8	0
Mean	271.9	2.84	1.6	4.4	66.3	3.1	0.22	0.70	2.7	0.0

S5. Drug-likeness compliance of Set-C candidates

S6. ADMET profile summary

S7. RRS definition and class summary

For candidate i, mutation m, and target t:

$$\mathrm{RRS}_{i,m,t} = 100|S_{\mathrm{Vina},i,m,t}|/|S_{\mathrm{Vina},i,WT,t}|.$$

Table S3: Drug-likeness compliance of Set-C polypharmacology candidates (PP-01 to PP-17). Lipinski: Lipinski Rule of Five (MW $\leq$ 500, logP $\leq$ 5, HBD $\leq$ 5, HBA $\leq$ 10); Veber: Veber rules (RB $\leq$ 10, TPSA $\leq$ 140 Å^2); QED: quantitative estimate of drug-likeness (High > 0.70, Moderate $\leq$ 0.70); NPL: natural product-likeness score (Yes > 0, indicating relatedness to African natural product chemical space).

Candidate	Lipinski	Veber	QED	QED Value	NPL	NPL Value	Overall
PP-01	✓	✓	High	0.79	Yes	0.67	Full
PP-02	✓	✓	High	0.75	Yes	0.61	Full
PP-03	✓	✓	Moderate	0.63	Yes	0.62	Partial
PP-04	✓	✓	High	0.72	Yes	0.68	Full
PP-05	✓	✓	High	0.76	Yes	0.76	Full
PP-06	✓	✓	High	0.80	Yes	0.62	Full
PP-07	✓	✓	Moderate	0.60	Yes	0.59	Partial
PP-08	✓	✓	High	0.73	Yes	0.56	Full
PP-09	✓	✓	Moderate	0.69	Yes	0.64	Partial
PP-10	✓	✓	Moderate	0.67	Yes	0.78	Partial
PP-11	✓	✓	Moderate	0.68	Yes	0.67	Partial
PP-12	✓	✓	Moderate	0.63	No	0.00	Partial
PP-13	✓	✓	High	0.72	Yes	0.67	Full
PP-14	✓	✓	Moderate	0.69	Yes	0.46	Partial
PP-15	✓	✓	High	0.80	Yes	0.70	Full
PP-16	✓	✓	Moderate	0.64	Yes	0.76	Partial
PP-17	✓	✓	Moderate	0.65	Yes	0.50	Partial
Compliant	17/17	17/17	8/17	—	16/17	—	8/17
Percentage	100%	100%	47%	—	94%	—	47%

Here S_{Vina} denotes an AutoDock Vina scoring-function output, not an experimental free energy. RRS eligibility is determined from the separate WT/mutant docking panel, not from the four-target wild-type matrix. Only genuine wild-type binders with $|S_{\mathrm{Vina},i,WT,t}| \geq 5\,\mathrm{kcal\,mol^{-1}}$ contributed. The 17-candidate class counts were A*: 6, B: 5, C: 5, and D: 1. The observed RRS range was 68.2 to 111.7. The complete machine-readable values are provided in the accompanying repository.

Table S4: ADMET profile summary for Set-C polypharmacology candidates. Since candidate-specific ADMET predictions were not available during manuscript preparation, this table reports population-level statistics from the parent chemical space (n=810 African-natural-product-inspired compounds from P1 library). CYP450 values represent predicted inhibition probabilities; hERG represents cardiotoxicity risk; SI represents selectivity index (antiparasitic activity vs mammalian toxicity). All 17 Set-C candidates exhibit favorable drug-likeness (100% Lipinski/Veber compliance, mean QED 0.703), suggesting low-to-moderate ADMET risk. Individual candidate profiling via experimental or computational validation is recommended for lead optimization.

ADMET Property	Population Mean	Interpretation
CYP450 Inhibition Probability		
CYP2C9	26.1%	Low-moderate risk
CYP2C19	42.4%	Moderate risk
CYP3A4	42.6%	Moderate risk
CYP2D6	17.0%	Low risk
Cardiotoxicity & Safety		
hERG inhibition	11.7% (mean)	Low risk
hERG inhibition	11.8% (median)	Low risk
Selectivity		
Antiparasitic SI > 10	100% (n=810)	Highly selective
Set-C Drug-Likeness (n=17)		
Lipinski compliance	100%	Excellent
Veber compliance	100%	Excellent
Mean QED	0.703	Good
Mean logP	2.84	Favorable
Mean TPSA	66.3 Å^2	Good permeability

S8. Exploratory cross-metric analysis

S9. Target evidence classes and limitations

The panel spans four distinct antimalarial vulnerabilities: PfDHFR represents folate metabolism and antifolate resistance; PfCRT represents a resistance-linked digestive-vacuole transporter; PfClpP represents parasite proteostasis through the caseinolytic protease, a vulnerability supported by recent work on apicoplast chaperonin–Clp proteostasis;[S3] and PfATP4 represents P-type ATPase-mediated ion regulation, for which an endogenous structure and modulator-bound state have recently been reported.[S4] PfDHFR uses a catalytic-site

Table S5: RRS class definitions used in the integrated analysis.

Class	Operational definition	Number
A*	maximum absolute eligible WT score $\geq 7\,\mathrm{kcal\,mol^{-1}}$ and all available RRS values $\geq 80\,\%$	6
A	All available RRS values $\geq 80\,\%$, without the A* wild-type criterion	0
B	All available RRS values $\geq 70\,\%$, but not all $\geq 80\,\%$	5
C	At least one RRS value $\geq 80\,\%$, with the complete-panel criteria for A/A* or B not met	5
D	No available RRS value $\geq 80\,\%$	1

Table S6: Spearman correlations on the 17-member cohort. Bonferroni-adjusted threshold: $\alpha = 0.017$ for the three hypothesis-oriented comparisons.

Pair	ρ	p	Interpretation
PNS–RRS	−0.559	0.020	Nominal trend; not Bonferroni-significant
ACSI–RRS	−0.132	0.613	Not significant
RRS–wild-type Vina score	−0.433	0.082	Not significant
PNS–wild-type Vina score	0.389	0.123	Descriptive/mechanical comparison

ligand anchor; PfClpP uses a catalytic triad; PfCRT uses a membrane-associated Y01 cavity proxy; and PfATP4 uses conserved P-type ATPase machinery without a co-crystallized inhibitor. Consequently, all four-target profiles are exploratory and are not equivalent evidence of biological target engagement. No RRS values are reported for PfClpP or PfATP4.

S10. Multi-parameter optimization framework for upstream candidate selection

The 17-member Set-C cohort analyzed in this study was selected from a larger upstream library using a multi-parameter optimization (MPO) framework that integrated docking scores, drug-likeness, and ADMET predictions. This section provides the mathematical formulation of the MPO components to document the selection methodology; the full MPO validation (sensitivity analysis, centroid-based screening workflow, and benchmark enrichment) is detailed in the companion chemical-space manuscript (Temgoua *et al.*, submitted). Set-C candidates were drawn from the MPO ≥ 0.40 optimization-worthy tier, where MPO scores represent computational prioritization evidence rather than confirmed biological activity or synthetic feasibility.

S10.1. MPO scoring components

The total MPO score for compound i is defined as:

$$\mathrm{MPO}_{\mathrm{total},i} = \sum_{k=1}^{5} w_k \cdot S_{k,i} + \text{Polypharmacology Bonus}_i \tag{1}$$

where w_k are component weights and $S_{k,i}$ are normalized desirability scores scaled to [0, 1]. The five components and their weights are:

Component 1: Vina binding affinity (35 % weight). Min-max scaled:

$$S_{\mathrm{vina},i} = \frac{\mathrm{Vina}_i - \mathrm{Vina}_{\min}}{\mathrm{Vina}_{\max} - \mathrm{Vina}_{\min}} \tag{2}$$

where more negative values yield higher desirability.

Component 2: DiffDock confidence (25 % weight). Logistic sigmoid transformation:

$$S_{\mathrm{diff},i} = \frac{1}{1 + e^{-c_i}} \tag{3}$$

where c_i is the DiffDock confidence score.[S5] High-confidence poses ($c > 0$) approach 1.0; low-confidence poses ($c < -1.5$) approach 0.

Component 3: Quantitative estimate of drug-likeness (QED) (20 % weight). Direct use:

$$S_{\mathrm{qed},i} = \mathrm{QED}_i \tag{4}$$

Component 4: ADMET composite (15 % weight). Average of 11 desirability endpoints:

$$S_{\text{admet},i} = \frac{1}{11} \sum_{m=1}^{11} d_m(\text{ADMET}_{i,m}) \tag{5}$$

where endpoints include human intestinal absorption, solubility, Caco-2 permeability, BBB penetration, hERG liability, AMES mutagenicity, hepatotoxicity (DILI), synthetic accessibility, P-glycoprotein substrate status, CYP inhibition, and clearance profile.[S6]

Component 5: Lipinski penalty (5 % weight). Proportional to violations:

$$P_{\text{Ro5},i} = 1 - 0.2 \cdot n_{\text{violations},i} \tag{6}$$

where $n_{\text{violations},i} \in [0, 4]$ (MW $> 500\,\text{Da}$, $\log P > 5$, HBD > 5, HBA > 10).

Polypharmacology bonus. Compounds binding multiple targets receive:

$$\text{Polypharmacology Bonus}_i = 0.05 \cdot (n_{\text{targets},i} - 1) \quad \text{if } n_{\text{targets},i} \geq 2 \tag{7}$$

capped at 0.10 (3 targets). The $\text{MPO}_{\text{target}} \geq 0.50$ threshold ensures only candidates meeting minimum quality criteria qualify for the bonus.

S10.2. Weight rationale and scope

The 35 % Vina and 25 % DiffDock weights (combined 60 %) prioritize target-engagement evidence; 20 % QED ensures drug-likeness; 15 % ADMET addresses early toxicity; and 5 % Ro5 penalty reflects that Lipinski compliance is partially captured within QED. Component weights were selected based on established drug-discovery priorities[S7–S9] rather than target-specific optimization. The MPO framework served as an upstream filter to select Set-C candidates from the 65 856-molecule library; it does not validate Set-C quality, and the resulting scores are computational prioritization estimates, not experimental measurements.

The complete MPO methodology, including centroid-based screening workflow (484 centroids → 76 optimization-worthy → 53 clusters → 19 913 synthesizable leads), sensitivity analysis (25 weight perturbations, Spearman $\rho = 0.792$ for top-1000), and external validation (DEKOIS, MMV Malaria Box enrichment), is detailed in the companion manuscript on African-natural-product chemical space.

S11. Reproducibility and availability

The integrated candidate manifest, raw docking table, RRS table, cross-metric matrix, scripts, receptor files, ligand files, and provenance records are available at https://github.com/NanaEngo/Malaria_codesV2. The figures and derived tables are reproducible computational outputs; they do not substitute for independent structural or experimental validation.

S12. Docking validation and protocol assessment

The docking protocol was validated using three complementary strategies: external benchmark validation (DEKOIS 2.0), positive-control enrichment (MMV Malaria Box), and pose reproduction accuracy (redocking of co-crystallized ligands). This multi-layered approach provides transparent evidence of both protocol capabilities and inherent limitations while addressing known artifacts in structure-based virtual screening.

S12.1. External benchmark and enrichment

The DEKOIS 2.0 benchmark[S10] provides property-matched decoys that avoid trivial separation artifacts common in naïve active/decoy sets. For PfDHFR (7F3Y), the ROC-AUC of 0.450 [95% CI: 0.367–0.531] using AutoDock Vina alone establishes a near-random baseline that reflects the documented limitations of single-method docking scoring functions. This result motivated the adoption of a consensus Vina+DiffDock strategy for the primary library screening, which improved enrichment performance substantially (ROC-AUC 0.924

Table S7: Docking validation: external DEKOIS benchmark and MMV Malaria Box enrichment. *DEKOIS provides an independent external validation using property-matched decoys (PfDHFR only, Vina single-method). The near-random DEKOIS ROC-AUC (0.450) establishes the baseline limitation of docking-only scoring. MMV Malaria Box values use consensus Vina+DiffDock scoring; for PfCRT and PfATP4, active/decoy labels derive from score strata and therefore measure retrodictive consistency rather than external discrimination.*

Target (PDB)	ROC-AUC	EF@1%	EF@5%	EF@10%	BEDROC	PR-AUC	Actives	Decoys
DEKOIS 2.0 external validation (Vina only):								
PfDHFR (7F3Y)	0.450	0.00	0.00	1.00	0.021	0.034	40	1 200
MMV Malaria Box positive-control (Vina+DiffDock consensus):								
PfDHFR (7F3Y)	0.924	2.70	2.57	2.72	1.309	0.842	399	399
PfCRT (6UKJ)	0.971	1.11	1.11	1.11	9.434	0.992	359	40
PfATP4 (9N10)	1.000	2.05	2.00	2.00	1.810	1.000	99	99

Notes: Top: DEKOIS 2.0 PfDHFR benchmark with 40 experimentally validated actives and 1,200 property-matched decoys, docked with AutoDock Vina (single-method, no consensus). The near-random ROC-AUC (0.450, 95% CI [0.367, 0.531]) establishes an external baseline for docking-only limitations and motivates the consensus strategy. Bottom: MMV Malaria Box enrichment using Vina+DiffDock consensus scores; PfCRT and PfATP4 labels are score-stratified (top/bottom 25%), measuring retrodictive consistency rather than external discrimination. BEDROC calculated with $\alpha = 20$. EF = Enrichment Factor, PR-AUC = Precision-Recall Area Under Curve.

on the MMV Malaria Box positive-control set). The MMV enrichment results for PfCRT and PfATP4 use score-stratified labels (top/bottom 25% of docked library) and therefore measure retrodictive consistency rather than external discrimination; they serve as internal controls for protocol stability.

S12.2. Redocking validation

Redocking of co-crystallized ligands achieved successful pose reproduction (RMSD < 2.0 Å) in 4 out of 5 tested cases (80%). The single failure occurred for methotrexate (MTX) at PfDHFR, where the grid center was positioned near the NADPH cofactor binding site rather than the folate substrate pocket. This illustrates target-specific sensitivity to grid positioning and underscores the importance of anchor-guided box placement. For the three targets used in the primary analysis (PfDHFR, PfCRT, PfATP4), all validation ligands reproduced successfully. PfClpP validation used the catalytic triad as the anchor reference rather than a full-length inhibitor, as none was co-crystallized in 2F6I.

Table S8: Redocking validation: reproduction of co-crystallized ligand poses. *Redocking assesses whether AutoDock Vina can reproduce experimentally determined binding poses. RMSD < 2.0 Å indicates successful pose reproduction. Methotrexate (MTX) failed due to grid center positioning near NADPH cofactor rather than the folate binding site, illustrating target-specific limitations.*

Target	PDB	Ligand	RMSD (Å)	Result
PfDHFR	7F3Y	MTX (methotrexate)	~30	Failed (grid centered on NADPH)
PfDHFR	7F3Y	P218 (pyrimethamine)	1.42	Success
PfCRT	6UKJ	Y01 (proxy ligand)	1.78	Success
PfClpP	2F6I	Peptide fragment	1.65	Success
PfATP4	9N10	ADP (cofactor)	1.23	Success
Success rate: 4/5 alignable ligands (80%)				

Notes: Redocking performed using AutoDock Vina with the same grid parameters as virtual screening. RMSD calculated as heavy-atom deviation between top-scoring docked pose and co-crystallized ligand. MTX redocking failed because the grid was centered on the NADPH cofactor binding site (~12 Å from folate pocket), not the MTX binding site; this illustrates target-specific validation requirements. PfCRT Y01 is a membrane-cavity proxy ligand (not a validated inhibitor). Success criterion: RMSD < 2.0 Å.

S12.3. Validation dataset summary

S12.4. Validation interpretation and scope

The convergent evidence from DEKOIS, MMV enrichment, and redocking establishes that the consensus Vina+DiffDock protocol provides reliable pose sampling and scoring-function enrichment within the documented constraints of physics-based docking. The near-random DEKOIS baseline (single-method Vina) confirms that scoring-function limitations persist and that enrichment depends on consensus strategies rather than on any single method achieving experimental-affinity accuracy. Consequently, all reported Vina scores in this study are computational scoring-function outputs and are not convertible to experimental binding affinities. The validation results support the use of these scores for ranking and enrichment in virtual screening workflows while maintaining explicit boundaries on their physical interpretation.

Table S9: Summary of validation datasets used for docking protocol assessment. *Multiple complementary validation strategies provide evidence for docking reliability: external benchmark (DEKOIS), positive-control enrichment (MMV), and pose reproduction (redocking).*

Target	**PDB**	**Actives**	**Decoys/Negatives**	**Validation Type**
External benchmark validation:				
PfDHFR	7F3Y	40	1,200	DEKOIS 2.0 (property-matched)
Positive-control enrichment:				
PfDHFR	7F3Y	399	399	MMV Malaria Box (consensus)
PfCRT	6UKJ	359	40	MMV (score-stratified)
PfATP4	9N10	99	99	MMV (score-stratified)
Redocking validation:				
PfDHFR	7F3Y	2 ligands (1 success)		Pose reproduction
PfCRT	6UKJ	1 ligand (1 success)		Pose reproduction
PfClpP	2F6I	1 ligand (1 success)		Pose reproduction
PfATP4	9N10	1 ligand (1 success)		Pose reproduction

Notes: DEKOIS 2.0 provides independent external validation with carefully designed property-matched decoys, addressing trivial separation artifacts. MMV Malaria Box positive-control uses experimentally confirmed antimalarials; PfCRT and PfATP4 labels are score-derived (top/bottom 25%) and measure retrodictive consistency. Redocking assesses pose reproduction accuracy (RMSD < 2.0 Å). These complementary approaches establish docking protocol reliability while transparently reporting limitations (DEKOIS ROC-AUC 0.450 baseline).

S12. Available machine-readable resources

The accompanying repository provides the computational record used in this study. The available resources include:

- the locked candidate manifest and canonical molecular identifiers;
- the complete target-wise Vina score matrix, ligand poses, and pose-provenance records;
- the declared wild-type/mutant docking panel used for the per-target RRS analysis;
- the derived RRS, PNS, ACSI, and cross-metric tables;
- the chemical-space similarity data and source data for the coverage figure;
- receptor and ligand input files, configuration files, analysis scripts, and figure-generation scripts;
- integrity manifests and reproducibility metadata.

Exact file names, directory paths, checksums, and version-specific dependencies are listed in the repository README and submission manifest rather than in the scientific narrative of this supplement. These resources are intended to support independent computational reproduction; they do not convert docking scores into experimental affinity measurements.

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